Henry Wang

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Web Infrastructure Engineer · Mini-Programs / Rendering / DevTools

SUMMARY

Web infrastructure engineer with 8 years of experience in frontend and platform engineering, specializing in mini-program engines, cross-platform rendering and developer tooling. Led two mini-program engines from zero to production at Kuaishou and Binance. Works across the full stack of the runtime — browser engines (Chromium / WebKit), compilers, rendering architecture and native mobile (iOS / Flutter) — with a recent focus on AI-driven developer productivity.

> **What is a mini-program?** An evolution of the mobile web: H5 apps that run inside a host mobile app, with logic and rendering split into separate threads and a virtual DOM driving the UI, combining near-native performance with the dynamic delivery of the web.

SKILLS

Frontend & Architecture	JavaScript / TypeScript, rendering architecture, template compilers, dynamic UI components
Engines & Debugging	Chromium source customization, iOS JSContext / WebKit, Chrome DevTools Protocol
Mobile	Objective-C, Swift, Flutter, native rendering integration
Games & AI	Phaser.js, Cocos engine integration; AI agent developer tools, dev-workflow automation

EXPERIENCE

Binance — Web Infrastructure Engineer / Mini-Program Lead

May 2021 – Nov 2026

- **Mini-program engine:** Led development of Binance's mini-program engine from scratch, owning the rendering architecture, API layer and compiler. Drove company-wide adoption and opened the platform to third-party merchants; mini-program pages reached **50%** of total Binance App DAU.
- **Engine re-architecture:** Initiated and led a core rewrite moving rendering from a template-based to a data-driven model, improving performance by **30%+** and enabling Flutter and native mobile rendering — the foundation for dynamic components.
- **Dynamic components:** Packaged mini-programs as embeddable cards, banners and toasts that can be placed anywhere in the app, powering personalized, per-user dynamic rendering.
- **Mini-game platform:** Built mini-game support from scratch by integrating Phaser.js with the mini-program rendering architecture; partnered with the Cocos engine team to launch GameFi titles on the Binance mini-game platform.
- **AI developer tools:** Built AI-driven developer tools for mini-programs, iOS and Android with an integrated agent, connecting app UI, logs and build pipelines so AI can assist across the entire development workflow.
- **Issue triage platform:** Unified client, backend, web and error-reporting systems and standardized in-app alerting; customer screenshots now map directly to the responsible code and owner, cutting resolution time.

Kuaishou — Web Infrastructure Engineer

Sep 2019 – May 2021

- **Mini-program engine:** Built Kuaishou's mini-program engine from scratch, responsible for the rendering architecture, APIs and compiler, through launch and third-party merchant onboarding. Tracked down an iOS JSContext engine bug (WebKit Bug 161929) to resolve rendering inconsistencies between iOS and Android.
- **Mini-program DevTools:** Built DevTools from scratch on the Chrome DevTools Protocol, connecting DevTools to mobile browsers to enable on-device debugging of mini-programs.
- **Performance metrics:** Modified Chromium source to expose internal performance metrics to page scripts, and built a service for measuring page performance.

Beijing Weishili Technology Co., Ltd. — Frontend / Mobile Engineer

Sep 2018 – Sep 2019

- Developed and maintained all of the company's web projects, including the internal admin console, creator management console and official website.
- Developed iOS apps with Objective-C, Swift and Flutter; shipped 2 apps to the App Store.

EDUCATION

Nankai University

Bachelor's Degree · 2014 – 2018